



Gaurav Soni

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B.Tech. in Computer Science

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EDUCATION

•Indian Institute of Information Technology, Vadodara

B.Tech in Computer Science (3rd Year) – CPI: 6.8

June 2023 – Present

•Shiv Jyoti Senior Secondary School

Class 12 (RBSE Board) – 90.2% (First Division)

April 2022 – March 2023

•Kendriya Vidyalaya, Deogarh (Rajasthan)

Class 11 (CBSE) – 84.6% | PCM with Computer Science

April 2011 – March 2022

•JEE Mains

97.7 Percentile

2023 - first attempt

EXPERIENCE

•HackIIITV 2024

Game Developer – Unreal Engine 5

October 2024

IIIT Vadodara

- Built a single-player thriller mystery game in a team of 3 using Unreal Engine 5, C++, and Blueprints.
- Handled visual appearance, color grading, and 3D object modelling; implemented audio-based navigation mechanics.
- Cleared 3 evaluation rounds among 10+ competing teams.

PERSONAL PROJECTS

•ZOMOREELS – Reel-Based Food Discovery Platform

Full-Stack Web Application (Dual-Role Architecture)

2025

- Built and deployed a REST API-driven full-stack web application with API integration for media delivery and user interactions food discovery app where restaurants upload short-form food reels and users browse a TikTok-style scroll-snap video feed with like, save, and bookmarked-reel tab — deployed on **Vercel + Render**.
- Architected a **dual-role auth system** (User & Food Partner) using JWT in **HttpOnly cookies** (7-day expiry with synced cookie **maxAge**); middleware distinguishes **TokenExpiredError** — clears stale cookie and returns structured 401 instead of a generic error.
- Implemented **Zod input validation** on all POST endpoints via a reusable middleware factory; applied **try/catch** across every controller with safe error responses and **environment-based CORS** for seamless local-to-production switching.
- Built end-to-end **media pipeline**: Multer (in-memory buffer) → base64 → **ImageKit CDN** (UUID filename) → CDN URL stored in MongoDB; like/save system uses Mongoose **.populate()** with atomic **\$inc** on **likeCount** and optimistic UI updates.
- Live: (click here) | Source: (click here) | **Stack**: React 19 (Vite), Node.js, Express, MongoDB, Mongoose, JWT, Zod, Multer, ImageKit

•Smart Agriculture Monitoring & Decision Support System

Full-Stack IoT Simulation & Crop Recommendation Platform

2024 – Present

- Developed a full-stack IoT-based soil monitoring platform with real-time dashboards and intelligent crop recommendations.
- Engineered a rule-based decision engine analyzing soil moisture, pH, temperature, humidity, and NPK levels to generate irrigation and fertilizer recommendations for 12+ crops.
- Designed MongoDB data models (StateProfile, CropRule, Reading) and built REST APIs using Node.js and Express.js, powering real-time dashboards with Chart.js visualizations.
- Live: View Project | Source: GitHub
- Tools & Technologies: React (Vite), Node.js, Express.js, MongoDB, Mongoose, Tailwind CSS, Chart.js.

•Personal Portfolio Website

Portfolio to showcase other projects and skills

2026

- Built a responsive developer portfolio using React and Vite to showcase projects and technical skills.
- Integrated EmailJS-based contact form for direct communication and deployed the application on Vercel for global access.
- Live: (click here) | Source code: (click here)
- Tools & Technologies: React, JavaScript, HTML5, CSS3, Vercel.

TECHNICAL SKILLS AND INTERESTS

Languages: C, C++, JavaScript

Developer Tools: Git, GitHub, Postman, NPM, Vercel

Frontend: React, Next.js, Tailwind CSS, HTML5, CSS3, Vite

Backend: Node.js, Express.js, REST APIs

Tools: Git, GitHub, Postman, NPM, Vercel, Render

Databases: MongoDB, MySQL

Soft Skills: Problem Solving, Analytical Thinking, Clean Code Practices, Communication, Teamwork

Coursework: Data Structures and Algorithms, Object Oriented Programming, Database Management Systems, Operating Systems

Areas of Interest: Backend Engineering, Full-Stack Applications, Secure Authentication

Coding Profiles: Click - TakeUForward (TUF) – Solving Striver's A2Z DSA Sheet and practicing Data Structures & Algorithms problems

POSITIONS OF RESPONSIBILITY

•**Project Leader – Smart Agriculture Systems**, Embedded Systems Project *Jan 2026 – Present*

- Led a team to design and develop a crop recommendation system based on location, soil type, and environmental parameters.
- Architected system logic to suggest primary and secondary crops along with suitable fertilizers and soil treatments.
- Coordinated hardware-software integration and managed overall project planning and execution.

EXTRA-CURRICULAR ACTIVITIES

•**Lead Guitarist, Encore – The Music Club** IIIT Vadodara *2023 – Present*

- Performed at multiple inter-college cultural fests including IIT Gandhinagar Battle of Bands.
- Secured 3rd position in an inter-college solo instrumental competition among multi-instrument participants.

•**Basketball Team Member** IIIT Vadodara *2023 – 2024*

- Represented the institute in inter-college tournaments and led the team in competitive fixtures.
- Played in multiple competitions including Concours (DAIICT Annual Sports Fest).